Project Design Phase – II

Solution Requirements (Functional & Non-Functional)

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Team ID: SWTID-2026-5429

Project Name: UCab – MERN Stack Cab Booking System

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Register using Name, Email and Password
		Validate user details before registration
		Store user information securely in MongoDB
FR-2	User Login	Login using registered Email and Password
		Authenticate user using JWT Token
		Redirect user to dashboard after successful login
FR-3	User Authentication	Encrypt passwords using bcryptjs
		Secure protected routes using JWT authentication
		Logout functionality
FR-4	Cab Booking	Select pickup location
		Select destination location
		Choose cab type
		Confirm booking
FR-5	Fare Estimation	Calculate estimated fare based on ride details
		Display estimated fare before booking confirmation

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-6	Booking Management	View booking details
		Update booking (if applicable)
		Cancel booking
FR-7	Ride History	Display previously booked rides
		View booking date and ride information
FR-8	User Profile Management	View profile
		Update personal information
		Logout from application
FR-9	Database Management	Store user information
		Store booking records
		Retrieve booking history
FR-10	Responsive User Interface	Support desktop, tablet and mobile devices
		Easy navigation across all pages

Non-Functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application provides a simple, intuitive, and responsive user interface for easy cab booking.
NFR-2	Security	User passwords are encrypted using bcryptjs, and JWT authentication is used to protect authorized routes and APIs.
NFR-3	Reliability	The system maintains consistent booking information and ensures reliable communication between frontend, backend, and database.

NFR No.	Non-Functional Requirement	Description
NFR-4	Performance	API responses are optimized for faster execution, and MongoDB queries are efficiently handled using Mongoose.
NFR-5	Availability	The application is accessible through a web browser and can be deployed on local or cloud servers for continuous availability.
NFR-6	Scalability	The MERN architecture supports future expansion such as Driver Module, Admin Dashboard, Live Tracking, and Online Payment Integration without major structural changes.
NFR-7	Maintainability	The project follows a modular folder structure with reusable React components and MVC architecture in Express.js for easier maintenance.
NFR-8	Compatibility	The application is compatible with modern web browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox.
NFR-9	Data Integrity	MongoDB ensures secure storage and retrieval of user and booking information while maintaining data consistency.
NFR-10	Responsiveness	The user interface automatically adapts to different screen sizes including desktops, tablets, and mobile devices using Bootstrap.